

Instructions: Write your answers to the following questions in a separate document, and upload it to D2L in PDF format. For full credit, you **must** show your work, i.e., you must show how you derived your answer, or explain your reasoning. You also cannot receive partial credit for an incorrect answer if you do not show any work.

1 (5 points). A computer's virtual memory system has a 32-bit address space and a 16-KiB page size. The page table, which uses 4 bytes per entry, is stored entirely in specialized hardware attached to the MMU. When a process starts, the page table is copied to the hardware from main memory, at one 32-bit word every 60ns. Suppose the OS is running processes using a round-robin scheduler with a quantum of 120 ms, which includes the time it takes to load the process's page table. What percentage of a process's quantum is spent loading its page table? What quantum length should the scheduler choose if it wants the overhead of loading the page table to be 10%?

2 (15 points). Suppose we have a computer with 512 GiB of physical memory, a 64-bit logical address space, and a page size of 32 KiB.

Part A (10 points). If we want to use a multilevel page table to address this machine's memory, how many levels will be needed so that the highest level page table fits within one page? Using that number of levels, how many bits of a virtual memory address should be used for each level of the page table? Describe the layout of the virtual memory address.

Part B (5 points). If, instead, we want to use an inverted page table, how large would the inverted page table be? How many pages would it occupy?